

JKBOSE 2026

CLASS 11TH

REDOX REACTIONS

LECTURE-01 CHEMISTRY

VISHAL SIR

Bhaichara
ON
TOP.



JKBose Wallah

Topics To Covered

- 1** **Concept of Oxidation and Reduction Reactions**
- 2** **Calculation of Oxidation Number**

REVISION

Buffer Solution:→

Whose pH remains constant.

- Acidic (Weak Acid & CB).
- Basic (WB & CA).

Henderson - Hasselbalch Equation



$$K = \frac{[H_3O^+][A^-]}{[HA][H_2O]} \text{ or } \underline{K \times [H_2O]} = \frac{[H_3O^+][A^-]}{[HA]}$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

$$[H_3O^+] = K_a \frac{[HA]}{[A^-]}$$

$$\log [H_3O^+] = \log \left(K_a \times \frac{[HA]}{[A^-]} \right)$$

$$-pH = \log K_a + \log \frac{[HA]}{[A^-]}$$

$$-pH = -pK_a + \log \frac{[HA]}{[A^-]}$$

$$pH = pK_a - \log \frac{[HA]}{[A^-]}$$

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

$$pH = pK_a + \log \frac{[CB]}{[WA]}$$

Applications of Buffer Solution

- (i) Normal blood has pH of 7.4. This pH is maintained by buffer
Buffer of carbonic acid (H_2CO_3 and NaHCO_3)
- (ii) Buffers find numerous applications in industrial processes, like manufacture of
paper, drugs, dyes, inks, paints etc.
- (iii) Buffer solutions are applied for digestion process of food.
- (iv) These are used as preservatives of foods and fruits.
- (v) Buffer can be used for preservation of many dairy and agricultural products.

Buffer Capacity

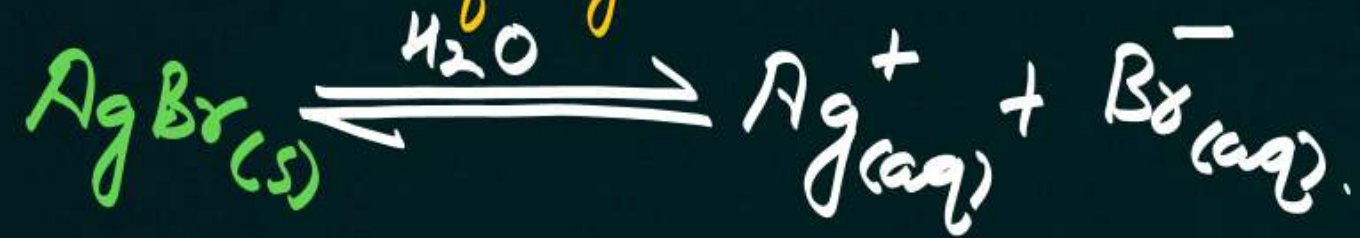
Buffer capacity is quantitatively defined as the number of moles of acid or base added in 1 L of solution to change the pH by unity.

$$\text{Buffer capacity } (\phi) = \frac{\text{Number of moles of acid or base added to 1 L of buffer}}{\text{Change in pH}}$$

Solubility Product (K_{sp})

Salts like AgBr , AgCl , BaSO_4 , PbCl_2 etc. dissolve in water to very small extent and those salt that dissolve in water negligibly are called sparingly soluble salts.

e.g. Dissolution of AgBr in water.



According to law of Chemical Equilibrium

$$K = \frac{[\text{Ag}^+][\text{Br}^-]}{[\text{AgBr}]}$$

But concentration of pure solid like AgBr is taken as constant.

$$K \times [\text{AgBr}] = [\text{Ag}^+][\text{Br}^-]$$

$$K_{sp} = [\text{Ag}^+][\text{Br}^-]$$

K_{sp} is a solubility product & is defined as product of concentration of ions of sparingly soluble salt.

Solubility of ions is determined by the concentration.

$$\text{i.e. } K_{sp} = [\text{Ag}^+][\text{Br}^-]$$

$$\text{or } K_{sp} = (S)(S)$$

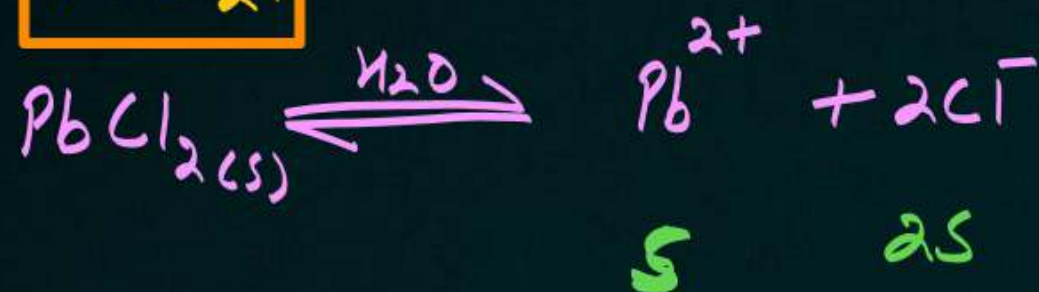
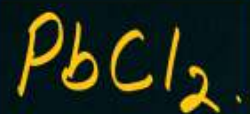
$$\text{or } K_{sp} = S^2$$

$$\text{or } S^2 = K_{sp} \text{ or } S = \sqrt{K_{sp}}$$

$$S = \sqrt{K_{sp}}$$

Solubility Product

Q. Determine the solubility (s).



$$K_{sp} = [\text{Pb}^{2+}][\text{Cl}^-]^2$$

$$K_{sp} = (s)(2s)^2$$

$$K_{sp} = s \times 4s^2$$

$$K_{sp} = 4s^3$$

$$\text{or } 4s^3 = K_{sp}$$

$$s^3 = \frac{K_{sp}}{4}$$

$$s = \left(\frac{K_{sp}}{4}\right)^{1/3}$$



$$K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}]$$

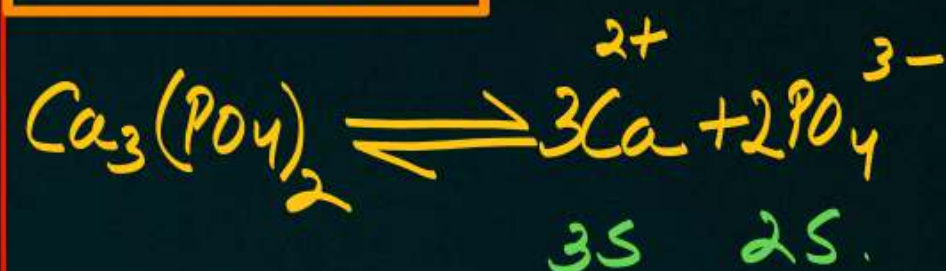
$$K_{sp} = s \times s$$

$$K_{sp} = s^2$$

$$(K_{sp})^{1/2} = s$$

$$s = (K_{sp})^{1/2}$$

$$s = \sqrt{K_{sp}}$$



$$K_{sp} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2$$

$$K_{sp} = (3s)^3 (2s)^2$$

$$K_{sp} = 27s^3 \times 4s^2$$

$$K_{sp} = 27 \times 4 \times s^3 \times s^2$$

$$K_{sp} = 108 s^5$$

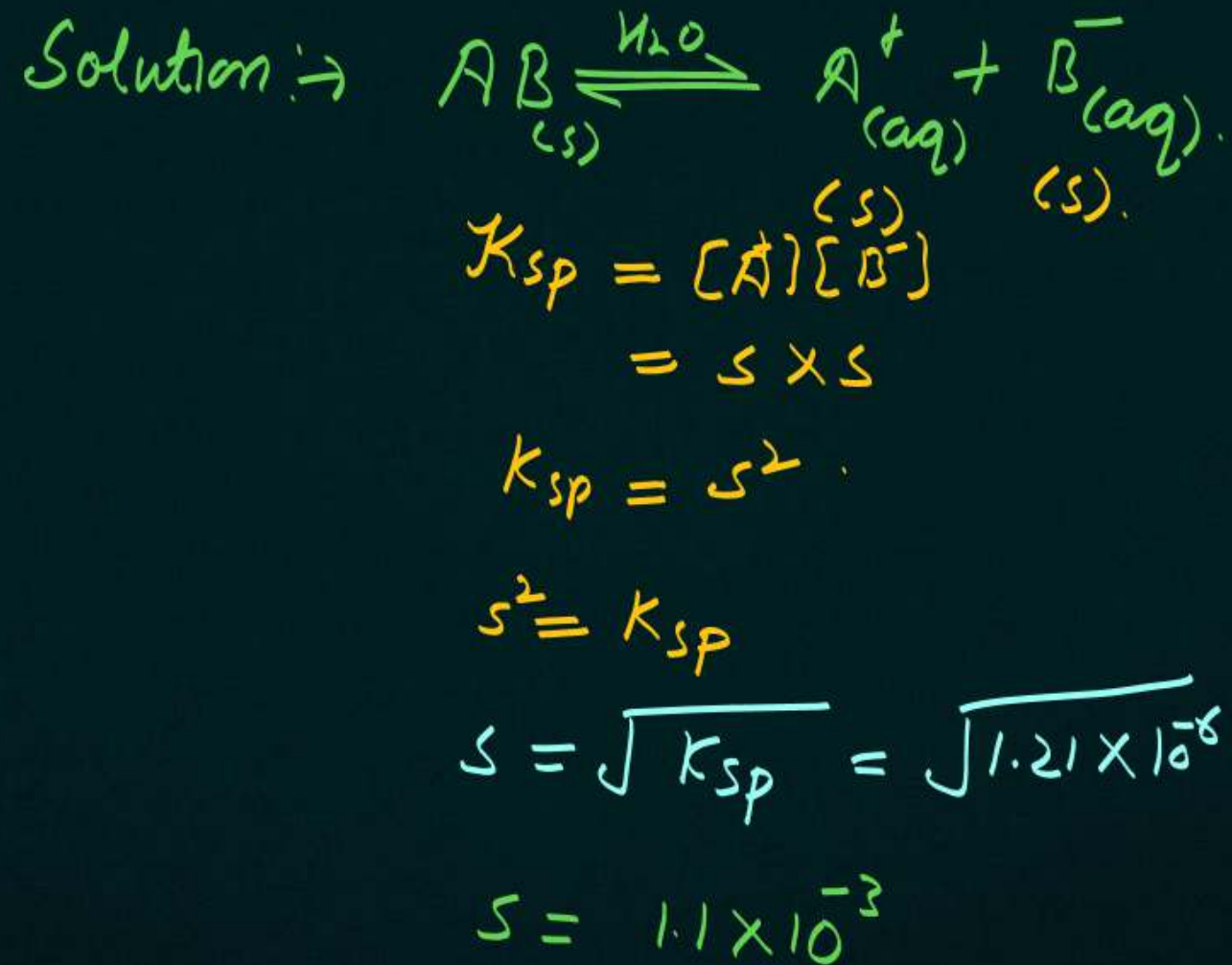
$$\frac{K_{sp}}{108} = s^5$$

$$\text{or } \left(\frac{K_{sp}}{108}\right)^{1/5} = s$$

QUESTION

The solubility product of a sparingly soluble salt AB at room temperature is 1.21×10^{-6} , its molar solubility is

- 1 $1.21 \times 10^{-6} \text{ M}$
- 2 $1.1 \times 10^{-4} \text{ M}$
- 3 $1.1 \times 10^{-3} \text{ M}$
- 4 None of these



SYLLABUS



UNIT-VII: REDOX REACTION

[4 MARKS]

- Concept of oxidation and reduction, redox reactions, oxidation number, balancing of chemical equations in redox reactions, applications of redox reactions.

Concept of Oxidation and Reduction Reactions

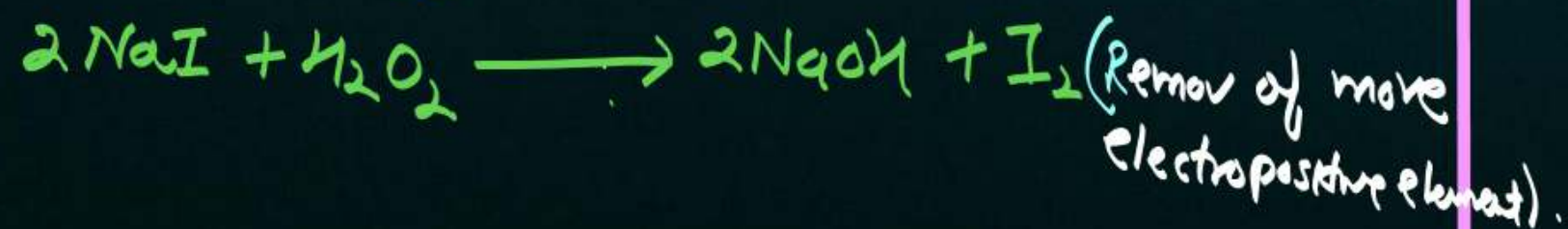
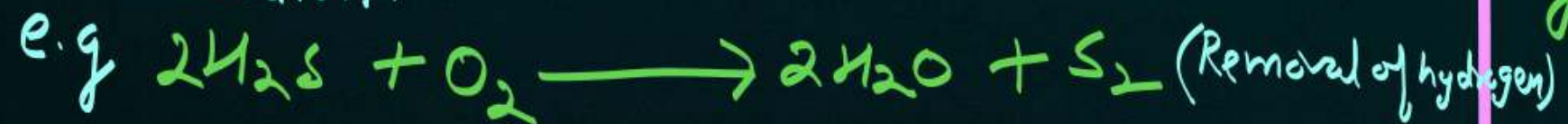


A) CLASSICAL CONCEPT.

- Oxidation $\rightarrow$ Process in which addition of oxygen or more electro-negative element is called oxidation.

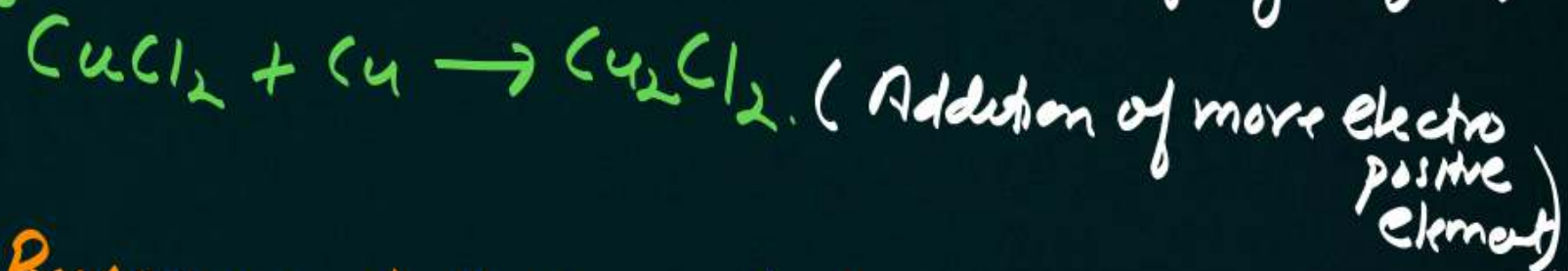


- Process in which removal of hydrogen or more electro positive element takes place is called oxidation.

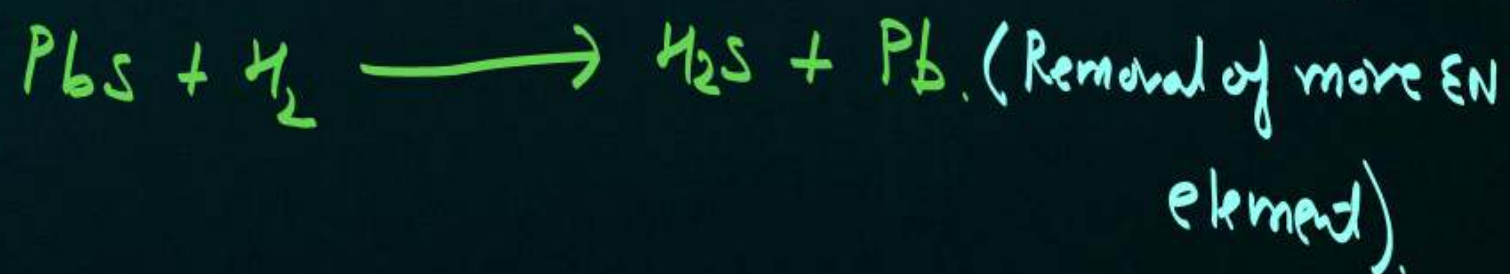


REDUCTION $\rightarrow$

Process in which addition of hydrogen or more electro positive element takes place is called Reduction.



- Process in which removal of oxygen or more electronegative element take place is called reduction.



Oxidation

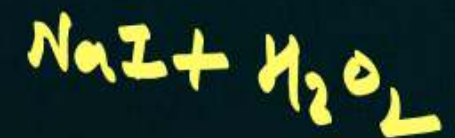
- Addition of oxygen or more EN element
- Removal of hydrogen or more electropositive element

Reduction

- Addition of hydrogen or more electropositive element
- Removal of oxygen or more EN element



Concept of Oxidation and Reduction Reactions



Concept of Oxidation and Reduction Reactions

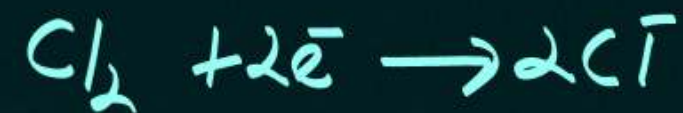
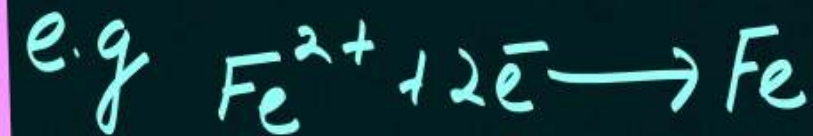
B) MODERN CONCEPT.

Oxidation $\rightarrow$ Reaction in which loss of e^- takes place is called oxidation.



During Oxidation increase in +ve charge or decrease in -ve charge takes place.

Reduction $\rightarrow$ Reaction in which gain of e^- takes place is called reduction.



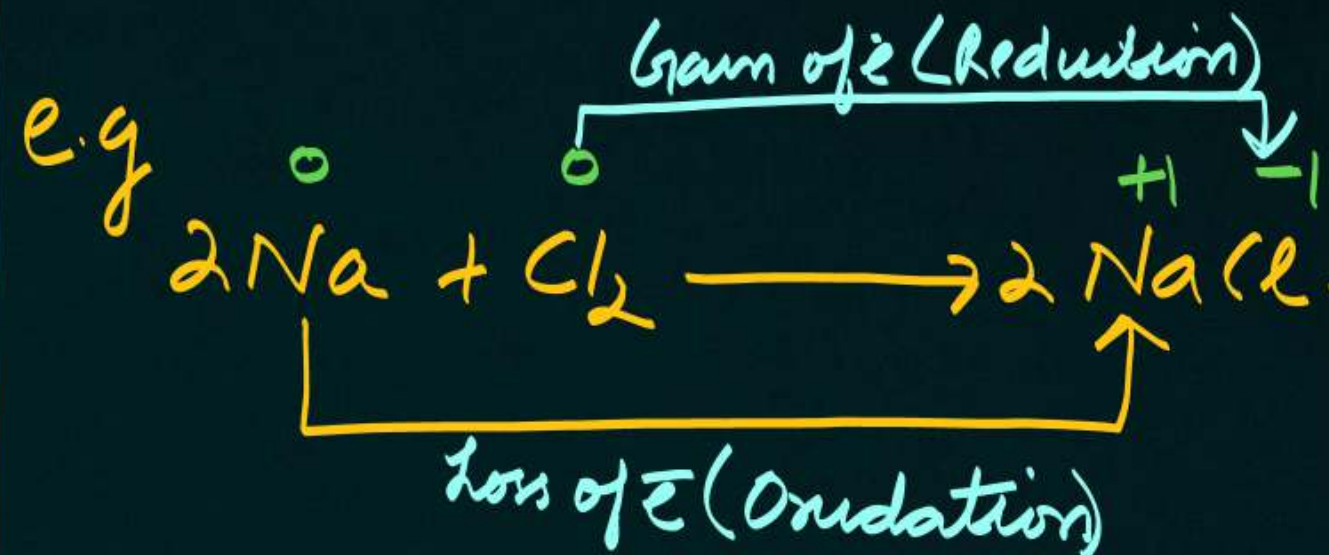
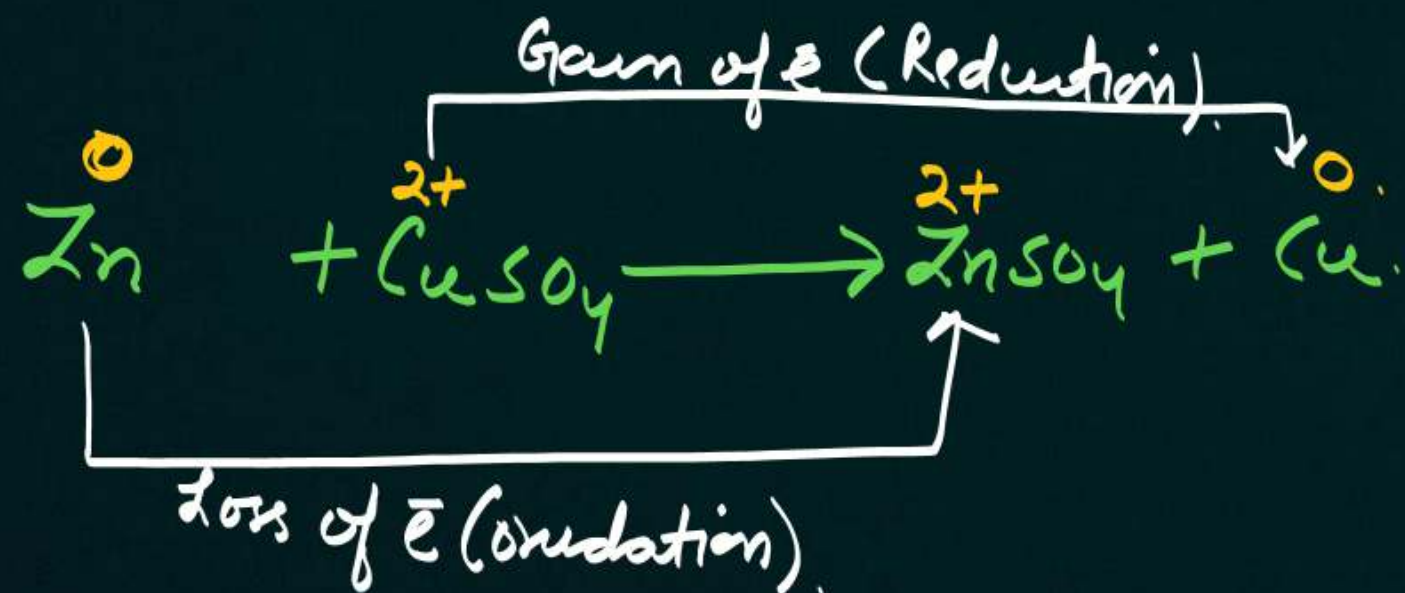
During reduction increase in -ve charge takes place or decrease in +ve charge takes place.

Concept of Oxidation and Reduction Reactions

Redox Reaction:→

Types of reaction in which reduction as well as oxidation takes place simultaneously is called redox reaction.

e.g.



Love You 

Thank
You

#XUV